

Suppositories & pessaries



Chapter Learning objectives

At the end of the session, students will be able to:

- Define suppositories and pessaries**
- Describe the desirable properties of bases, classification of bases, other excipients used**
- Explain general procedures used for the preparation of suppositories**
- Calculate displacement value, amount of base and drug required in suppository preparation**
- Describe packaging, labeling and storage requirements of suppositories**

Suppositories

- Suppositories are **solid dosage** forms intended for insertion into body orifices where they **melt, soften, or dissolve** and exert local or systemic effects.
- Pessaries are suppositories for insertion into vaginal cavity.
 - Pessaries are nowadays called **vaginal inserts**.

Suppositories.....

- After insertion, suppositories **melt ,soften,** or **dissolve** in the cavity fluid.
- Drugs having systemic effects such as **sedatives, tranquilizers, analgesics** and others can be administered by rectal suppositories.
- Sites of use of suppositories:
 - Commonly used rectally
 - Used some extent vaginally
 - Occasionally used urethrally

Suppositories.....

❑ Size and shapes

- Rectal suppositories are usually about **32 mm long**, are **cylindrical**, and have **one or both ends tapered**.



Suppositories.....

- Some rectal suppositories are shaped like a **bullet, a torpedo, cone-shaped or the little finger.**



Suppositories.....

- Adult rectal suppositories are about 2 g when cocoa butter is used as a base.
- Depending on density of base and medicament the weight may vary.
- Rectal suppositories for infants and children are half the weight and size of adult suppositories.

Suppositories.....

- Pessaries usually are **globular or oviform** and sometimes **cone** shaped like rectal suppositories.
- They are 5 g in weight when cocoa butter is used as a base.



Suppositories.....

- Urethral suppositories (bougies) are **thin, cylindrical** in shape (**pencil-shaped**).
- for male :
 - ✓ diameter: 3 to 6 mm
 - ✓ length: ~ 140 mm
 - ✓ Weight: 4 g
- for female:
 - ✓ length: ~ 70 mm (half of male)
 - ✓ Weight: 2 g (half of male)

Suppository bases

Selection of suppository/pessary bases

- The ideal suppository base should meet the following general specifications:
 - ✓ The base is **non toxic** and **non irritating** to mucous membranes.
 - ✓ The base is **compatible** with variety of drugs.
 - ✓ The base **melts or dissolves** in rectal fluids.
 - ✓ The base should be **stable** on storage.

Classification of suppository bases

- Suppository bases can be classified broadly into three types:
 - ✓ **Fatty or oleaginous bases**
 - ✓ **Water-soluble or water miscible bases**
 - ✓ **miscellaneous bases**
- In addition to the bases, the following ingredients may be added: **surface-active agents, preservatives** and **coloring agents**.

Classification of suppository bases...

- **Fatty or oleaginous bases**

- ✓ These type of bases **melt or soften at body temperature**. Example, Cocoa butter (theobroma oil) and semi-synthetic fats

- **Water-soluble or water miscible bases**

- ✓ These type of bases **dissolve or disperse** in body fluid. E.g., glycleo-gelatin bases and PEG bases

- **Miscellaneous bases**

- ✓ mixtures of oleaginous and water-soluble materials.

Cocoa butter (theobroma oil)



- This has been used for over 200 years.
- It is a natural vegetable fat obtained from roasted seeds of *theobroma cacao* (*cacao tree*). A plant which is cultivated in tropical countries.

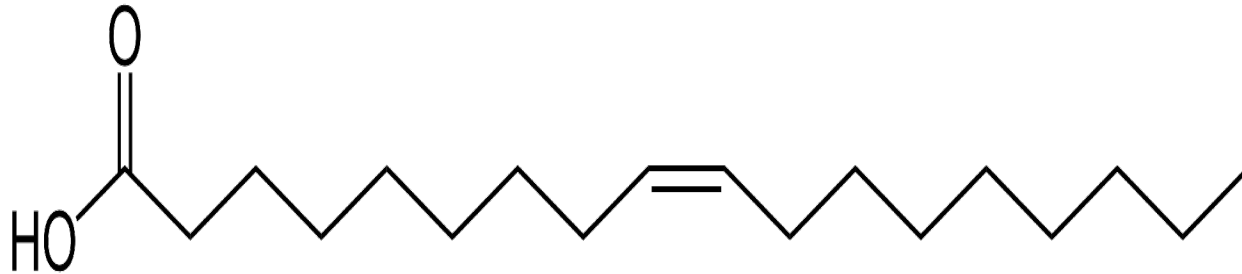


Cocoa butter (theobroma oil).....

- It is a yellowish-white solid with chocolate like odour.
- It is **triglyceride** (combination of glycerin and one or different fatty acids). The fatty acids involved in it are **stearic acid, palmitic acid, oleic acid** and others.

Cocoa butter (theobroma oil).....

- Oleic acid is the main constituent and accounts 1/3 of all fatty acids.
- Chemical structure of oleic acid.



Cocoa butter (theobroma oil).....

■ Valuable features of cacao butter:

- ✓ Melting point range of 30 to 36 °C; hence it is solid at normal room temperature but melts in the body.
- ✓ Ready liquefaction on warming and rapid settling on cooling.
- ✓ Miscibility with many ingredients.

Cocoa butter (theobroma oil).....

❑ Limitations of cacao butter:

- **Polymorphism** (i.e., its ability to exist in more than one crystalline forms).
- heat treatment of cocoa butter is a maximum temperature of 40 to 50 °C to avoid change in crystal structure and melting point.

Cocoa butter (theobroma oil).....

- If overheated, e.g., 60°C and cooled rapidly, the suppository formed will melt at a temperature below 30 °C.
- if **overheated and cooled rapidly**, it produces, on cooling, unstable
 - **alpha crystals**, melting at about **20 °C** or
 - **gamma crystals**, which melt at about **15 °C**,

Cocoa butter (theobroma oil).....

- It will take several days for the stable crystalline structure to reform to give a stable product at room temperature.
- If melted at **not more than 40-50 °C** and slowly cooled it forms stable **beta crystals** with normal melting point.

Cocoa butter (theobroma oil).....

- The unstable forms eventually return to the stable condition but this may take several days.
- Meanwhile, the suppository may not set at room temperature or, if set by cooling, may re melt in the warmth of the patient's home.

Cocoa butter (theobroma oil).....

- This lowering of the solidification point can also lead to **sedimentation of suspended solids.**
- Great care must be taken to avoid overheating the base when making theobroma oil supps.

Cocoa butter (theobroma oil).....

➤ Adherence to mold

- ✓ does not contract enough on cooling.
- ✓ Sticking may occur, particularly if the plate is worn
- Sticking can be prevented by lubricating the mould before use with **glycerol** or **propylene glycol** or other liquid.
- The lubricant forms **a thin film** b/n the plate of the mould and suppository base.

Cocoa butter (theobroma oil).....

- The lubricant should be immiscible with the base (if miscible with the base, the film will be destroyed).
- For water miscible bases, mineral oil can be used as a lubricant.

Cocoa butter (theobroma oil).....

- **Softening point too low for hot climate**
- ✓ Refrigeration is required in warmer climate.
- ✓ To raise the softening point, **white beeswax (4%)** may be added to cocoa butter suppositories intended for use in tropical and subtropical countries.

Cocoa butter (theobroma oil).....

➤ Melting point may be reduced by soluble ingredients

- ✓ Certain drugs will depress the melting point of theobroma oil. Eg. Chloral hydrate and phenol.
- ✓ This effect can be countered by addition of a higher melting wax, such as **white bees wax**.

Cocoa butter (theobroma oil).....

➤ **Slow deterioration during storage**

- This is due to oxidation of the unsaturated glycerides.

➤ **Poor water-absorbing capacity**

- This can be improved by the addition of W/O emulsifying agents

➤ **It is immiscible with body fluids**

- This may inhibit the diffusion of fat-soluble drugs.

Cocoa butter (theobroma oil).....

➤ Leakage from the body

- Sometimes melted base escape from rectum or vagina.
- This is most troublesome with pessaries, because of their large size and therefore pessaries are rarely made from cocoa butter.

➤ Relatively high cost

Hydrogenated vegetable oils (Semi-synthetic fats)

- These are hydrogenated and subsequently heat treated products of vegetable oils such as **palm kernel oil, arachis oil, coconut oil, etc.**

Hydrogenated vegetable oils (Semi-synthetic fats).....

- The semi-synthetic fatty bases are made as follows:
 - ✓ First, the vegetable oils are hydrolysed to get free fatty acids;
 - ✓ Second, hydrogenation of the free fatty acids will be made;
 - ✓ Last, re-estriifying the fatty acid by heating with glycerol.

Hydrogenated vegetable oils (Semi-synthetic fats)...

- ✓ Hydrogenation saturates unsaturated glycerides and hydrolyses splits some of the triglycerides into fatty acids, glycerol and partial esters (mono- and di-glycerides).

Hydrogenated vegetable oils (Semi-synthetic fats)...

- ✓ Controlled modification of the process yields a wide range of materials containing mixtures of mon-, di-, and triglycerides of the saturated fatty acid (fatty acids C_9 - C_{18}).
- ✓ Examples of proprietary fats:
 - Wecobee[®], Suppocire[®], Massupol[®], Dehydag[®], Coberine[®], Witepsol[®]

Hydrogenated vegetable oils (Semi-synthetic fats)...

- ❑ **Advantages of hydrogenated vegetable oils over cocoa butter**
 - ✓ They have good resistance to oxidation because their unsaturated fatty acids have been reduced.
 - ✓ The difference between melting and setting point is small; generally only 1.5 to 2 °C and seldom over 3 °C.
 - Hence, they set quickly, the risk of sedimentation is low.

Hydrogenated vegetable oils (Semi-synthetic fats)...

- ✓ They usually contain a proportion of **partial glycerides**, e.g., glyceryl monostearates are W/O emulsifying agents and therefore, their emulsifying and water-absorbing capacities are good.

Hydrogenated vegetable oils (Semi-synthetic fats)...

- ✓ No mould lubricant is needed because they **contract significantly** upon cooling .
- ✓ They produce white, clean and almost odorless suppositories.

Hydrogenated vegetable oils (Semi-synthetic fats)...

❑ Disadvantages:

- ✓ Softening point is low for hot climate
 - Refrigeration is required
- ✓ They should not be cooled in refrigerator or ice because they become brittle if cooled quickly.
 - Polysorbate 80 help to correct this problem.
- ✓ Polymorphic transition is possible

Hydrogenated vegetable oils (Semi-synthetic fats)...

- ✓ They are *less viscous than cocoa butter* when melted and at this stage sedimentation is greater.
- Thickeners such as **magnesium stearate, bentonite and colloidal silicone dioxide** may be added to increase the viscosity.

Water miscible suppository bases

- Water miscible suppository bases are of comparatively recent origin.
- There are two types of water-soluble or water miscible bases
 - ✓ **Glycero-gelatin/ glycerinated gelatin bases**
 - ✓ **PEG bases**



Glycero-gelatin/glycerinated gelatin bases

- ✓ Are mixture of glycerine, gelatin and water.
- ✓ Gelatin is a denatured protein obtained by partial hydrolyses of collagen.
- ✓ Collagen is obtained from animal skin or bone.
- ✓ Gelatin makes gel system rigid.

Glycero-gelatin base.....

□ Glycerol suppository BP 1988

Gelatin.....14 g

glycerol.....70 g

Purified Water, ad.....100 g

- Sometimes we can increase gelatin up to 20 g in case of hot climate (b/e the gel softens at a high temperature).
 - Also counteract the softening effect of any liquid ingredients included.

Glycero-gelatin base.....

❑ Application

- ✓ Frequently used for preparation of **vaginal supps.**
 - no leakage.
- ✓ Also used for formulation of urethral supps. (**gelatin 60%; glycerol 20%** and **water 20%**).
 - Cocoa butter is not used in urethral supps. b/e it brittles easily.

Glycero-gelatin base.....

❑ Disadvantages

- ✓ They have physiological effect.
 - Glycerine has laxative effect.
 - tend to absorb mucosal moisture and bring irritation of mucosal surface.
- ✓ They are hygroscopic (b/e of glycerol).
 - Suppositories must be stored in tightly closed glass containers.
- ✓ Require moistening with water before insertion.

Glycero-gelatin base.....

- ✓ Microbial contamination is likely (b/e of gelatin)
 - ✓ Preservatives like parabens are used if the suppository is intended for extended shelf-life.
- ✓ Suppository preparation is time consuming.

Glycero-gelatin base.....

- ✓ Removal of formed suppositories is difficult from mould
 - they do not contract significantly.
 - lubrication is essential with mineral oil.
- ✓ Slow dissolution of suppositories in cavity fluid.

PEGs/Macrogols

- In addition to their application in the formulation of ointments, PEGs are suitable as suppository or pessary bases.
- Mixture of two or more grades are used (i.e., low and high melting point ingredients).
- For Individual PEGs, as molecular weight increases, their hardness and melting point increases.
- The proportion of the PEGs governs the physical properties of the supps.

PEGs/Macrogols.....

- High molecular weight polymers give products that **dissolve and release their drugs slowly** and supps. that are **hard** and **brittle**.
- Softer, less brittle preparations that disperse and liberate their drug more quickly are obtained by mixing high with either medium or high with low polymers or by adding plasticizers .

Examples of PEG/macrogol bases

Type	Base I	Base II	Base III	Base IV	Base V
PEG 400	-	-	20	-	-
PEG 1000	-	-	-	-	75
PEG 1540	-	33	33	94	-
PEG 4000	33	-	-	-	25
PEG 6000	47	47	47	-	-
Water	20	20	-	-	-
Hexane 1,2,6 triol (plasticizer)	-	-	-	6	-

PEGs/Macrogols.....

❑ Advantages

- ✓ The mixture generally has a melting point above 42 °C.
 - cool storage is not required
 - satisfactorily used in hot climates
- ✓ They do not stick to the mould and,
 - contract significantly on cooling,
 - lubricant is not required

PEGs/Macrogols.....

- ✓ produce solutions of high viscosity
 - leakage from body is not a problem.
- ✓ Supps. have a clean and smooth appearance.
- ✓ do not have laxative effect.
 - good for preparation of rectal supps.
- ✓ Microbial contamination is less likely.

PEGs/Macrogols.....

❑ Disadvantages

- ✓ Hygroscopic
- ✓ Poor bioavailability of some medicaments
 - The good solvent property of the base makes retention of the drug in the base → decreased absorption → reduced therapeutic effect
- ✓ Brittle
 - Addition of surfactant or plasticizer reduces this effect.

PEGs/Macrogols.....

- ✓ Crystal growth of certain medicaments may occur
 - particularly common for drugs partly in solution and partly in suspension in the base.
 - This makes the product brittle, Irritating and take longer to dissolve.
- ✓ Requires moistening with water before insertion (if the percentage of water is below 20%) to avoid irritation of mucous membrane.

PEGs/Macrogols.....

✓ Incompatibility

- Incompatible with bismuth salts, tannins, phenols
 - Liquefaction occur in the last two instances
- They lower activity of certain antibacterial agents. e.g., quaternary ammonium cpds and *P*-hydroxybenzoic esters (parabens)
- Dissolve certain plastics (e.g., polythene and Bakelite)
- Interacts with polystyrene vials.

Preparation of suppositories